Hardware components of Computer- Types and classifications:

Computer hardware consists of various components, including input/output devices, the central processing unit (CPU), motherboard, memory, and secondary storage devices. These components are classified into internal and external hardware. Internal hardware includes components that are installed within the computer's case, such as the CPU and motherboard, while external hardware includes devices that are connected to the computer, such as a keyboard or a printer.

Input/ Output devices and their classification:

Input devices are used to enter data into the computer, such as a keyboard or mouse, while output devices are used to display or print data, such as a monitor or printer. Input/output devices, such as a touch screen, allow for both input and output.

Eras of Business computing:

The eras of business computing include the mainframe era, the personal computer era, and the mobile computing era. The mainframe era was characterized by large, centralized computers, while the personal computer era saw the rise of individual computers. The mobile computing era is defined by the increasing use of smartphones and other mobile devices.

Client- Server computing:

Client-server computing refers to a model of computing in which a client computer requests resources or services from a server computer. This model is used in many applications, including email and web browsing.

System unit/CPU:

The system unit is the main part of a desktop computer that contains the CPU, memory, and other components. The CPU is the "brain" of the computer and performs most of the computer's processing tasks.

Motherboard:

The motherboard is the main circuit board of a computer that connects all of the other components, such as the CPU, memory, and input/output devices.

Computer memory - Characteristics, advantages and disadvantages:

Computer memory is used to temporarily store data that the computer is currently using. There are several types of computer memory, including random access memory (RAM) and read-only memory (ROM). The advantages of computer memory include its speed and its ability to store large amounts of data. Disadvantages include its volatility, meaning that data is lost when the computer is turned off, and its cost.

Secondary storage devices and their types:

Secondary storage devices are used to store data permanently, such as a hard drive or a flash drive. These devices can store much larger amounts of data than computer memory, but are slower to access.

Categories of computer - Varieties and examples:

There are many categories of computers, including personal computers, laptops, tablets, and smartphones. Each type of computer is designed for a specific purpose and has its own unique features and capabilities. For example, personal computers are designed for general-purpose computing, while smartphones are designed for mobile computing and communication.

Motherboard bridges

Motherboard bridges are components on a computer motherboard that allow different parts of the system to communicate with each other. There are typically two types of bridges found on a motherboard: Northbridge and Southbridge.

The Northbridge is responsible for communication between the CPU (Central Processing Unit), RAM (Random Access Memory), and the graphics card. It is connected directly to the CPU and allows for fast data transfer between the CPU and these components.

The Southbridge is responsible for communication between the other components of the system, such as the hard drive, USB ports, and other peripherals. It also manages lower speed connections to the CPU, such as the PCI bus and the LPC (Low Pin Count) bus.

In modern computer architecture, the distinction between the Northbridge and Southbridge has become less clear, as many of the functions that used to be handled by the Northbridge have been integrated into the CPU or the Southbridge. However, the concept of a bridge is still important in understanding how different components of a computer communicate with each other.

BIOS, or Basic Input/Output System

USB, in full universal serial bus

LCD stand for Liquid-crystal display

LED stand for light emitting diode

CRT stand for cathode ray tube

The general transformation cycle for information is **data to information to knowledge**.

The basic architecture of computer was developed by J. von Neumann,

The central processing unit (CPU) consists of six main components:

- control unit (CU)
- arithmetic logic unit (ALU)
- registers.
- cache.
- buses.
- clock.
    - Control Unit (CU): The Control Unit is responsible for controlling the flow of data and instructions between the CPU, memory, and other input/output devices. It interprets program instructions and coordinates the execution of program commands.
    - Arithmetic Logic Unit (ALU): The Arithmetic Logic Unit is responsible for performing mathematical and logical operations, such as addition, subtraction, multiplication, and division. It also performs logical operations such as AND, OR, NOT, and XOR.
    - Registers: Registers are small, high-speed storage areas within the CPU that hold data and instructions temporarily. They are used to store operands, memory addresses, and intermediate results during processing.
    - Cache: Cache is a small, high-speed memory area that is used to store frequently accessed data and instructions. It is designed to reduce the time needed to access data from main memory, improving system performance.
    - Buses: Buses are the pathways that allow data to flow between the CPU, memory, and other input/output devices. They consist of a collection of wires that carry information between components.
    - Clock: The Clock is a timing device that regulates the speed at which the CPU operates. It sends out a series of pulses that synchronize the various operations of the CPU, ensuring that they occur at the right time.

First Generation (1940s-1950s): The first computers were built using vacuum tubes and were large and expensive. They were mainly used for scientific and military purposes.

Second Generation (1950s-1960s): The second generation of computers used transistors instead of vacuum tubes, which made them smaller, more reliable, and cheaper. They also used punched cards and magnetic tape for data storage.

Third Generation (1960s-1970s): The third generation of computers used integrated circuits, which allowed for even smaller and faster computers. They also introduced magnetic disks for data storage.

Fourth Generation (1970s-1980s): The fourth generation of computers saw the development of microprocessors, which allowed for even more powerful and efficient computers. Personal computers became available during this time, and computer networking also began to develop.

Fifth Generation (1980s-present): The fifth generation of computers is characterized by the use of artificial intelligence and expert systems. It also includes the development of parallel processing, which allows for multiple processors to work together to solve complex problems.

A modem is a device that connects a computer or other devices to the internet. It takes the digital signal from the computer and converts it into an analog signal that can be transmitted over a telephone line, cable line, or fiber optic line. The modem at the other end of the connection then converts the analog signal back into a digital signal that can be understood by the receiving device. Essentially, the modem acts as a translator between the digital signals used by computers and the analog signals used by the internet infrastructure.

Router is a device that connects multiple devices to a network and allows them to communicate with each other. It also provides a way to connect the network to the internet. The router takes the data from the devices connected to the network and sends it to the appropriate destination. It also provides security features, such as firewalls and network address translation (NAT), to protect the network from unauthorized access.

In some cases, a modem and a router may be combined into a single device, known as a modem-router or gateway. This device provides both the translation function of a modem and the network connectivity and security features of a router.

In summary, a modem is a device that connects a computer or other device to the internet, while a router is a device that connects multiple devices to a network and provides network connectivity and security features.

Intranet: An intranet is a private computer network used within an organization. It is only accessible to the organization's employees, members, or other authorized personnel. The purpose of an intranet is to provide a secure platform for employees to share information, collaborate on projects, and access company resources. An intranet may include features such as email, messaging, file sharing, and online forms.

Extranet: An extranet is a private network that allows authorized external parties, such as suppliers, vendors, partners, or customers, to access a company's information or systems. An extranet can be used for collaboration, supply chain management, customer support, or other business processes. Unlike the public internet, an extranet is secured by firewalls, encryption, and other security measures to protect the company's data.

- while intranets are designed for internal use only, extranets provide a secure way to share information with external parties without compromising security or confidentiality.
- Both types of networks are important tools for modern businesses to improve communication, collaboration, and productivity

The general transformation cycle for information is a process that involves several stages to convert raw data into meaningful and useful information.

Input: The first stage of the transformation cycle is input, where raw data is collected from various sources, such as databases, sensors, forms, or other data entry methods. The input can be in various formats, including text, numbers, images, or audio.

Processing: The second stage is processing, where the collected data is transformed into a meaningful form. The processing can involve several operations, such as sorting, filtering, aggregating, calculating, or analyzing the data. The processing can be automated using software, algorithms, or artificial intelligence.

Output: The third stage is output, where the processed information is presented in a useful form. The output can be in various formats, such as reports, dashboards, graphs, charts, or other visualizations. The output can be customized according to the user's needs or preferences.

Feedback: The fourth stage is feedback, where the output is evaluated and used to improve the input or processing. The feedback can be in the form of user feedback, performance metrics, or other measures of effectiveness. The feedback can be used to refine the transformation cycle and improve the quality and relevance of the output.

Storage: The final stage is storage, where the processed information is stored for future use. The storage can be in various forms, such as databases, cloud storage, or other data repositories. The storage can be secured and protected against unauthorized access or loss.

In summary, the general transformation cycle for information involves input, processing, output, feedback, and storage, which together convert raw data into meaningful and useful information for decision-making, analysis, or other purposes.

A computer network is a group of interconnected computers and other devices that can share resources and communicate with each other. The purpose of a computer network is to allow devices to communicate and exchange information, such as data, files, and applications, through a shared communication medium.

There are several types of computer networks, including:

- Local Area Network (LAN): A LAN is a network that covers a small area, such as an office, building, or campus. It typically uses wired or wireless connections to connect computers and other devices.

- Wide Area Network (WAN): A WAN is a network that covers a larger geographic area, such as a city, country, or the world. It typically uses the internet or other long-distance connections to connect computers and other devices.

- Wireless Network: A wireless network is a network that uses wireless connections, such as Wi-Fi, to connect computers and other devices.

- Cloud Network: A cloud network is a network that uses cloud-based services to store and manage data and applications. It allows users to access data and applications from anywhere with an internet connection.

Computer networks can be used for various purposes, including:

- Sharing resources, such as printers, files, and applications, among computers and devices.
- Facilitating communication, such as email, instant messaging, and video conferencing.
- Providing internet access to multiple devices.
- Enabling remote access to data and applications.
- Improving efficiency and productivity by streamlining workflows and reducing redundant tasks.

There are three main types of cloud computing services:

- I**nfrastructure as a Service (IaaS)**: IaaS provides users with virtualized computing resources, such as servers, storage, and networks, on a pay-as-you-go basis. The users can configure and manage these resources according to their needs, without having to worry about the underlying infrastructure.

- **Platform as a Service (PaaS**): PaaS provides users with a platform for developing, deploying, and managing their own applications. The platform includes the necessary tools and services, such as programming languages, databases, and web servers, to build and run the applications.

- **Software as a Service (SaaS):** SaaS provides users with access to software applications over the internet, on a subscription basis. The users can access and use the applications without having to install or maintain any software on their own devices.